

Advanced (Habanero)**Open Ended Questions (FRQ)**

- Q9.** In a given week, a factory produces x units of Product A and y units of Product B. The production of Product A requires 3 hours of labor and 2 hours of machine time per unit. The production of Product B requires 2 hours of labor and 3 hours of machine time per unit. The factory has 480 hours of labor available per week and 600 hours of machine time. Write a system of equations that represents the number of units of Product A and Product B that can be produced.
- Q10.** A theater has 300 seats and plans to sell tickets at two price levels: 20 and 30. If they want to make at least 6,000 in revenue and sell at least 100 tickets at the 30 price level, write a system of inequalities to represent the situation. Solve the system and explain the solutions in context.

Chef's Tips

- Tip 1: Provide clear instructions and examples for students to follow.
- Tip 2: Make sure to review the learning objectives before administering the test.
- Tip 3: Encourage students to show their work and explain their reasoning.